System Architecture - Dated ETF Allocation Disclosures (fully synthetic; Python + Flask + SQLite)

UI Layer - src/ui/app.py (Shahmeer; pending)

Flask application
Routes / HTML templates / forms
search, detail, portfolio, create, update, delete
No SQL written here

Synthetic Retained Evidence - data/raw/

Fictional capture files (placeholder text, .pdf)
identities on reserved example.invalid domain
not real issuer documents

request with parameters

file-existence check

Business Logic - validation and orchestration

schema_validation.py (Taimoor)
validated_transaction around every write
ContextKey, parent minimums, Complete totals (0.005)
retained-file existence check

dicts / domain errors

raises on invalid state

validated function calls

retained references

Domain errors
ValidationError / NotFoundError
BusinessRuleError

Data Access Layer - src/dal (Mason)

repositories.py
8 parameterized functions
search / detail / entries / portfolio
create / update / batch / delete

acquires connection

seed/query entry

dict results

database.py
Database context manager
PRAGMA foreign_keys = ON (verified)
explicit BEGIN / COMMIT / ROLLBACK

seed_data.py + run_analysis_queries.py
validated synthetic seed, six analytical queries

parameterized SQL

schema + synthetic seed

rows sqlite3.Row

Persistence

SQLite database (data/etf_allocations.sqlite3)
12 STRICT tables, 13 foreign keys, 4 triggers